

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : E & TC Semester :VI Subject Code & Name: BTETPE604C Power Electronics Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain two transistor analogy of an SCR. Derive an expression for anode current in terms of transistor parameters?		06
B)	Draw and explain characteristics of IGBT.		06
C)	List commutation methods of SCR. Explain Class- A commutation.		06
Q.2	Solve Any Two of the following.		12
A)	With the help of circuit diagram and wave forms explain working of single phase semi-converter with RL load.		06
B)	With the help of circuit diagram and wave forms explain working of single phase full bridge converter with R load.		06
C)	With the help of circuit diagram and wave forms explain working of single phase semi-converter with R load.		06
Q. 3	Solve Any two of the following.		12
A)	Explain operation of set-down chopper. Derive an expression for its output voltage.		06
B)	List different control techniques for chopper. Explain any one.		06
C)	Explain operation of Type D chopper with the help of circuit diagram.		06
Q.4	Solve Any Two of the following.		12
A)	With the help of suitable diagram explain operation of PWM inverters.		06
B)	Explain the operation of single phase full bridge inverter.		06
C)	Write a note on voltage control and harmonic reduction in voltage source inverter.		06

Q. 5	Solve Any two of the following.		12
A)	Explain Series Loaded half bridge dc-dc converter.		06
B)	With the help of block diagram, explain On-Line UPS system.		06
C)	Explain concept of forward converters for SMPS.		06
	*** End ***		

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